

A-Z Linear Algebra & Calculus for AI, Data Science and Machine Learning

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Calculus for Machine Learning

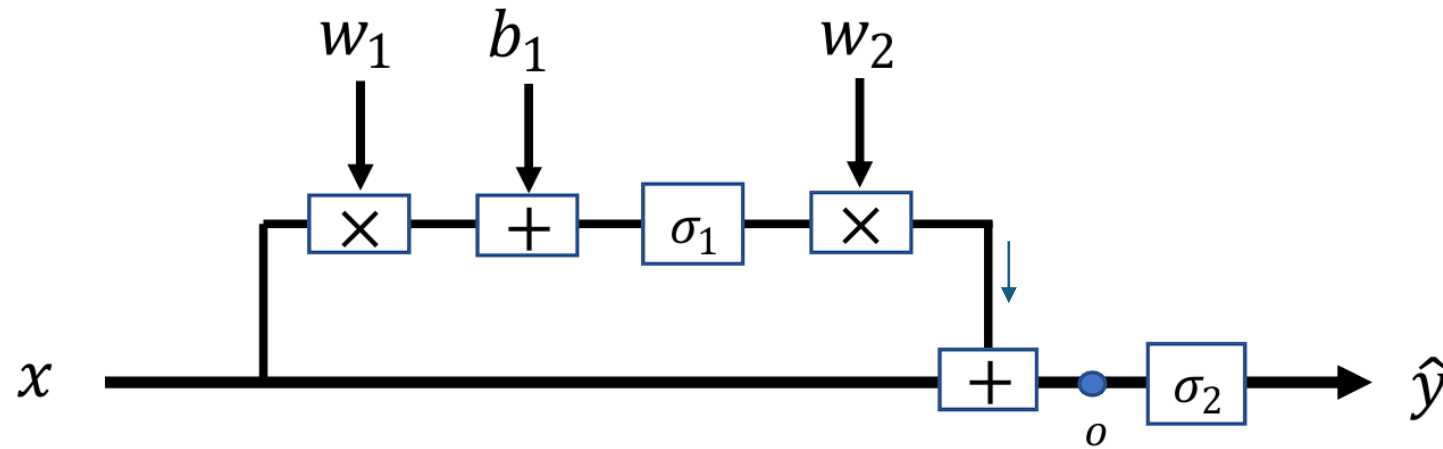
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Calculus is a branch of mathematics that studies **how things change**. It helps us understand motion, growth, **rates of change**, and areas under curves—stuff that's always shifting and evolving. There are two main types:

- **Differential Calculus** focuses on the rate of change, such as how fast a car goes at a specific moment (instantaneous speed). It's about slopes and derivatives. Understand how fast something is changing. **Example:** If you're driving, differential calculus tells you your speed at a specific second.
- **Integral Calculus** – focuses on the accumulation of quantities. Find totals from small pieces. **For example,** if you know your speed over time, integral calculus tells you how far you've traveled.

Why It Matters (Especially for Machine Learning)?

- Calculus is behind the scenes in training ML models.
- We use derivatives to adjust weights in neural networks (via gradient descent).
- It's also used in optimization, probability, and understanding model behavior.



Let $\sigma'(x)$ be the derivative of $\sigma(x)$, which are defined as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

$$\text{Loss} = \|\hat{y} - y\|_2^2 = (\hat{y} - y)^2$$

Solution:

$$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$$

$$\frac{\partial L}{\partial o} = \frac{\partial L}{\partial \hat{y}} \sigma'_2(x + w_2 \sigma_1(w_1 x + b_1))$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial o} \sigma_1(w_1 x + b_1)$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial o} w_2 \sigma'_1(w_1 x + b_1)$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial o} w_2 \sigma'_1(w_1 x + b_1) x$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial o} + \frac{\partial L}{\partial o} w_2 \sigma'_1(w_1 x + b_1) w_1$$

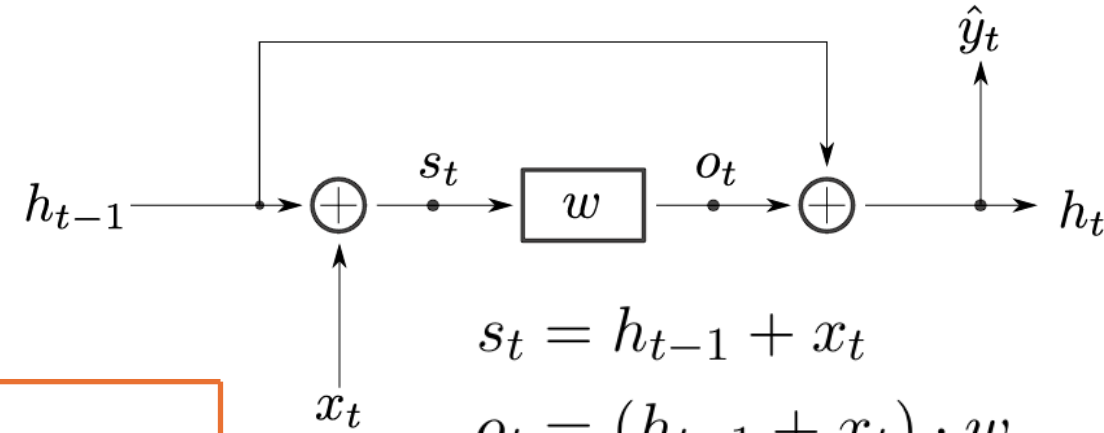
$$\rightarrow \frac{\partial L}{\partial \hat{y}_t} = 2(\hat{y}_t - y_t)$$

$$\rightarrow \frac{\partial L}{\partial o_t} = \frac{\partial L}{\partial \hat{y}_t} \cdot \frac{\partial \hat{y}_t}{\partial o_t} + \frac{\partial L}{\partial h_t} \cdot \frac{\partial h_t}{\partial o_t} = \frac{\partial L}{\partial \hat{y}_t} + \frac{\partial L}{\partial h_t}$$

$$\rightarrow \frac{\partial L}{\partial w_t} = \frac{\partial L}{\partial o_t} \cdot \frac{\partial o_t}{\partial w_t} = \frac{\partial L}{\partial o_t} \cdot (h_{t-1} + x_t) = s_t \cdot \frac{\partial L}{\partial o_t}$$

$$\rightarrow \frac{\partial L}{\partial w} = \sum_t \frac{\partial L}{\partial w_t}$$

$$\rightarrow \frac{\partial L}{\partial s_t} = \frac{\partial L}{\partial o_t} \cdot \frac{\partial o_t}{\partial s_t} = \frac{\partial L}{\partial o_t} \cdot w$$



$$s_t = h_{t-1} + x_t$$

$$o_t = (h_{t-1} + x_t) \cdot w$$

$$\hat{y}_t = h_{t-1} + o_t = h_t$$

$$\rightarrow \frac{\partial L}{\partial x_t} = \frac{\partial L}{\partial s_t} \cdot \frac{\partial s_t}{\partial x_t} = \frac{\partial L}{\partial s_t}$$

$$\rightarrow \frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial o_t} + \frac{\partial L}{\partial s_t}$$

If you have a function $f(x)$, the **derivative** is written as:

- $f'(x) \leftarrow$ (Lagrange notation)
- $\frac{df}{dx} \leftarrow$ (Leibniz notation)
- $Df(x) \leftarrow$ (less common, operator notation)

They all mean: how much does $f(x)$ change as x changes?

Let's say:

$$f(x) = x^2$$

Then the derivative is:

- $f'(x) = 2x$
- Or: $\frac{d}{dx}(x^2) = 2x$

If $f(x, y)$, then:

- $\frac{\partial f}{\partial x}$ = derivative with respect to x
- $\frac{\partial f}{\partial y}$ = derivative with respect to y

Used a lot in ML when you have functions with many variables.

$$f(x, y) = 3x^2y + 2xy^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(3x^2y + 2xy^2)$$

$$\frac{\partial f}{\partial x} = 6xy + 2y^2$$

- **First-Order** Partial Derivatives • $\frac{\partial f}{\partial x}$ • $\frac{\partial f}{\partial y}$

They show how the function changes as each variable changes on its own.

- **Second-Order** Partial Derivatives: These measure how the rate of change itself changes.
 - Pure Second-Order Derivatives • $\frac{\partial^2 f}{\partial x^2}$ • $\frac{\partial^2 f}{\partial y^2}$
 - Mixed Second-Order Derivatives • $\frac{\partial^2 f}{\partial x \partial y}$ • $\frac{\partial^2 f}{\partial y \partial x}$

For ML, Deep Learning, Optimization: $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

◆ Indefinite Integral (No Limits)

Represents the *antiderivative* (reverse of differentiation):

- $\int f(x) dx$

This gives a general function $F(x)$ such that $F'(x) = f(x)$

◆ Definite Integral (With Limits)

Represents the *area under the curve* from a to b :

- $\int_a^b f(x) dx$

This gives a number, not a function.

If $f(x) = 2x$:

- Indefinite integral:

$$\int 2x dx = x^2 + C \text{ (C = constant)}$$

- Definite integral:

$$\int_0^2 2x dx = [x^2]_0^2 = 4 - 0 = 4$$

Differential Calculus Basic Rules (Part-01)

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Rules in Partial Derivatives

- 1.Constant Rule**
- 2.Power Rule**
- 3.Sum and Difference Rule**
- 4.Product Rule**
- 5.Quotient Rule**
- 6.Chain Rule**
- 7.Mixed Partial Derivative Rule (Clairaut's Theorem)**
- 8.Higher-Order Derivatives Rule**
- 9.Implicit Differentiation Rule**

1. Power Rule

For a function $f(x, y) = x^n$ or $f(x, y) = y^n$:

$$\frac{\partial}{\partial x}(x^n) = nx^{n-1}$$

$$\frac{\partial}{\partial y}(y^n) = ny^{n-1}$$

Example:

$$f(x, y) = x^3$$

$$\frac{\partial f}{\partial x} = 3x^2$$

$$g(x, y) = y^4$$

$$\frac{\partial g}{\partial y} = 4y^3$$

2. Constant Multiple Rule

For a function $f(x, y) = c \cdot g(x, y)$, where c is a constant:

$$\frac{\partial}{\partial x}(c \cdot g(x, y)) = c \cdot \frac{\partial g(x, y)}{\partial x}$$

$$\frac{\partial}{\partial y}(c \cdot g(x, y)) = c \cdot \frac{\partial g(x, y)}{\partial y}$$

Example:

$$f(x, y) = 5x^2$$

$$\frac{\partial f}{\partial x} = 5 \cdot 2x = 10x$$

$$g(x, y) = 7y^3$$

$$\frac{\partial g}{\partial y} = 7 \cdot 3y^2 = 21y^2$$

3. Sum Rule

For a function $f(x, y) = g(x, y) + h(x, y)$:

$$\frac{\partial}{\partial x}(g(x, y) + h(x, y)) = \frac{\partial g(x, y)}{\partial x} + \frac{\partial h(x, y)}{\partial x}$$
$$\frac{\partial}{\partial y}(g(x, y) + h(x, y)) = \frac{\partial g(x, y)}{\partial y} + \frac{\partial h(x, y)}{\partial y}$$

Example:

$$f(x, y) = x^2 + y^2$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial x}(y^2) = 2x + 0 = 2x$$

$$g(x, y) = 3x^3 + 4y$$

$$\frac{\partial g}{\partial y} = \frac{\partial}{\partial y}(3x^3) + \frac{\partial}{\partial y}(4y) = 0 + 4 = 4$$

4. Product Rule

For a function $f(x, y) = g(x, y) \cdot h(x, y)$:

$$\frac{\partial}{\partial x}(g(x, y) \cdot h(x, y)) = g(x, y) \cdot \frac{\partial h(x, y)}{\partial x} + h(x, y) \cdot \frac{\partial g(x, y)}{\partial x}$$
$$\frac{\partial}{\partial y}(g(x, y) \cdot h(x, y)) = g(x, y) \cdot \frac{\partial h(x, y)}{\partial y} + h(x, y) \cdot \frac{\partial g(x, y)}{\partial y}$$

Simplified,

$$\frac{\partial f}{\partial x} = u \cdot \frac{\partial v}{\partial x} + v \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial f}{\partial x} = u \cdot \left(\frac{\partial}{\partial x} v \right) + v \cdot \left(\frac{\partial}{\partial x} u \right)$$

Example:

$$f(x, y) = (x^2) \cdot (y^3)$$

$$\frac{\partial f}{\partial x} = (x^2) \cdot \frac{\partial (y^3)}{\partial x} + (y^3) \cdot \frac{\partial (x^2)}{\partial x} = (x^2) \cdot 0 + (y^3) \cdot (2x) = 2xy^3$$

$$g(x, y) = (3x) \cdot (4y^2)$$

$$\frac{\partial g}{\partial y} = (3x) \cdot \frac{\partial (4y^2)}{\partial y} + (4y^2) \cdot \frac{\partial (3x)}{\partial y} = (3x) \cdot (8y) + (4y^2) \cdot 0 = 24xy$$

[More Examples => Next Slide](#)

Partial Derivative with Respect to x , $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

First, we compute the partial derivatives of $u(x, y) = x^2 + y^2$ and $v(x, y) = x^2y + y^2x$ with respect to x :

$$u(x, y) = x^2 + y^2$$

$$v(x, y) = x^2y + y^2x$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(x^2y + y^2x) = 2xy + y^2$$

$$\frac{\partial f}{\partial x} = u \cdot \frac{\partial v}{\partial x} + v \cdot \frac{\partial u}{\partial x}$$

$$\frac{\partial f}{\partial x} = (x^2 + y^2)(2xy + y^2) + (x^2y + y^2x)(2x)$$

Another approach:

$$f(x, y) = x^2(x^2y + y^2x) + y^2(x^2y + y^2x)$$

$$f(x, y) = x^4y + x^3y^2 + x^2y^3 + xy^4$$

$$\frac{\partial f}{\partial x} = 4x^3y + 3x^2y^2 + 2xy^3 + y^4$$

Final Answer:

$$4x^3y + 3x^2y^2 + 2xy^3 + y^4$$

Partial Derivative with Respect to y , $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

Now, we compute the partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to y :

$$\frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}(x^2y + y^2x) = x^2 + 2yx$$

Applying the product rule:

$$\frac{\partial f}{\partial y} = u \cdot \frac{\partial v}{\partial y} + v \cdot \frac{\partial u}{\partial y}$$

$$\frac{\partial f}{\partial y} = (x^2 + y^2)(x^2 + 2yx) + (x^2y + y^2x)(2y)$$

Final Answer:

$$\frac{\partial f}{\partial y} = x^4 + 2x^3y + 3x^2y^2 + 4xy^3$$

Partial Derivative with Respect to y , $f(x, y) = (x^2 + y^2)(x^2y + y^2x)$

Now, we compute the partial derivatives of $u(x, y)$ and $v(x, y)$ with respect to y :

Multiply both expressions:

$$f(x, y) = (x^2)(x^2y) + (x^2)(y^2x) + (y^2)(x^2y) + (y^2)(y^2x)$$

$$\frac{\partial f}{\partial y} = x^4 + 2x^3y + 3x^2y^2 + 4xy^3$$

(Final Answer)

Differential Calculus Basic Rules (Part-02)

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5. Quotient Rule

For a function $f(x, y) = \frac{g(x, y)}{h(x, y)}$:

$$\frac{\partial}{\partial x} \left(\frac{g(x, y)}{h(x, y)} \right) = \frac{h(x, y) \cdot \frac{\partial g(x, y)}{\partial x} - g(x, y) \cdot \frac{\partial h(x, y)}{\partial x}}{[h(x, y)]^2}$$

$$\frac{\partial}{\partial y} \left(\frac{g(x, y)}{h(x, y)} \right) = \frac{h(x, y) \cdot \frac{\partial g(x, y)}{\partial y} - g(x, y) \cdot \frac{\partial h(x, y)}{\partial y}}{[h(x, y)]^2}$$

Example:

$$f(x, y) = \frac{x^2}{y}$$

$$\frac{\partial f}{\partial x} = \frac{y \cdot \frac{\partial (x^2)}{\partial x} - x^2 \cdot \frac{\partial y}{\partial x}}{y^2} = \frac{y \cdot (2x) - x^2 \cdot 0}{y^2} = \frac{2xy}{y^2} = \frac{2x}{y}$$

$$g(x, y) = \frac{y^2}{x}$$

$$\frac{\partial g}{\partial y} = \frac{x \cdot \frac{\partial (y^2)}{\partial y} - y^2 \cdot \frac{\partial x}{\partial y}}{x^2} = \frac{x \cdot (2y) - y^2 \cdot 0}{x^2} = \frac{2xy}{x^2} = \frac{2y}{x}$$

Simplified,

$$f(x, y) = \frac{u(x, y)}{v(x, y)}$$

$$\frac{\partial f}{\partial x} = \frac{v \cdot \frac{\partial u}{\partial x} - u \cdot \frac{\partial v}{\partial x}}{v^2}$$

$$\frac{\partial f}{\partial x} = \frac{v \cdot \left(\frac{\partial}{\partial x} u \right) - u \cdot \left(\frac{\partial}{\partial x} v \right)}{v^2}$$

6. Chain Rule

If, $f(x) = h(g(x)) \rightarrow$ Then, $f'(x) = h'(g(x)) \cdot g'(x)$

$$f(x) = (3x + 2)^4 \rightarrow$$

- Outer: $h(u) = u^4$
- Inner: $u = g(x) = 3x + 2$

$$\frac{df}{dx} = h'(g(x)) \cdot g'(x) = 4(3x + 2)^3 \cdot 3 = 12(3x + 2)^3$$

Example:

$$f(x, y) = e^{x^2+y^2}$$

Let $u = x^2 + y^2$, then $f = e^u$.

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial f}{\partial u} = e^u$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} = e^{x^2+y^2} \cdot 2x = 2xe^{x^2+y^2}$$

$$= e^{x^2+y^2} \cdot \frac{d}{dx}(x^2 + y^2) = e^{x^2+y^2} \cdot (2x)$$

$$\boxed{\frac{\partial f}{\partial x} = 2x \cdot e^{x^2+y^2}}$$

Example: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - y_{\text{pred}})^2$ **Here,** $y_{\text{pred}} = mx_i + c$

Plug into the cost function: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2$

Take the derivative of Cost w.r.t. m :

$$\begin{aligned} \frac{\partial \text{Cost}}{\partial m} &= \frac{\partial}{\partial m} \left[\frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2 \right] \\ &= \frac{1}{n} \sum_{i=0}^n 2(y_i - (mx_i + c))(-x_i) \\ &= \frac{-2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}}) \end{aligned}$$

Final Answer: $\frac{\partial \text{Cost}}{\partial m} = -\frac{2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}})$

Example: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - y_{\text{pred}})^2$

Here, $y_{\text{pred}} = mx_i + c$

Plug into the cost function: $\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i - (mx_i + c))^2$

$$= y_i^2 - 2y_i(mx_i + c) + (mx_i + c)^2$$

$$\text{Cost} = \frac{1}{n} \sum_{i=0}^n (y_i^2 - 2y_i mx_i - 2y_i c + m^2 x_i^2 + 2mx_i c + c^2)$$

Take the derivative of Cost w.r.t. m : $\frac{d(\text{Cost})}{dm} = \frac{1}{n} \sum_{i=0}^n (-2y_i x_i + 2mx_i^2 + 2cx_i)$

$$= \frac{2}{n} \sum_{i=0}^n x_i (y_{\text{pred}} - y_i) \quad \left| \quad y_{\text{pred}} = mx_i + c \right.$$

Final Answer: $= -\frac{2}{n} \sum_{i=0}^n x_i (y_i - y_{\text{pred}})$

7. Mixed Partial Rule (Clairaut's Theorem)

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

Example:

$$f(x, y) = x^2 y^3 + xy$$

$$\frac{\partial f}{\partial x} = 2xy^3 + y$$

$$\frac{\partial^2 f}{\partial y \partial x} = 6xy^2 + 1$$

$$\frac{\partial f}{\partial y} = 3x^2 y^2 + x$$

$$\frac{\partial^2 f}{\partial x \partial y} = 6xy^2 + 1$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 6xy^2 + 1 \quad \text{(Final Answer)}$$

Differential Calculus

Some Important Calculus Formulas

Derivative

$$\frac{d}{dx} n = 0$$

$$\frac{d}{dx} x = 1$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} n^x = n^x \ln n$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

Integral (Antiderivative)

$$\int 0 \, dx = C$$

$$\int 1 \, dx = x + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{1}{x} \, dx = \ln x + C$$

$$\int n^x \, dx = \frac{n^x}{\ln n} + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arccot} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \operatorname{arccsc} x = -\frac{1}{x\sqrt{x^2-1}}$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \sec x \, dx = \sec x + C$$

$$\int \cot x \csc x \, dx = -\csc x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int -\frac{1}{\sqrt{1-x^2}} \, dx = \arccos x + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C$$

$$\int -\frac{1}{1+x^2} \, dx = \operatorname{arccot} x + C$$

$$\int \frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arcsec} x + C$$

$$\int -\frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arccsc} x + C$$

Partial Derivatives of Cost Functions in Machine Learning

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1. Mean Squared Error (MSE):

$$J = \frac{1}{n} \sum (\hat{y} - y)^2$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{2}{n} (\hat{y} - y)$$

In the context of the MSE & MAE and most machine learning models:

- y is called the **true value**, **actual value**, or **ground truth**.
- $\hat{y}$ (pronounced "y-hat") is called the **predicted value** or **model output**.

2. Mean Absolute Error (MAE):

$$J = \frac{1}{n} \sum |\hat{y} - y|$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{1}{n} \cdot \begin{cases} 1, & \hat{y} > y \\ -1, & \hat{y} < y \\ 0, & \hat{y} = y \end{cases}$$

Note:

The derivative of $|\hat{y} - y|$ depends on the **sign** of $\hat{y} - y$:

- If $\hat{y} > y$, then $|\hat{y} - y| = \hat{y} - y$ and the derivative w.r.t. $\hat{y}$ is 1.
- If $\hat{y} < y$, then $|\hat{y} - y| = -(\hat{y} - y) = y - \hat{y}$ and the derivative w.r.t. $\hat{y}$ is -1.
- If $\hat{y} = y$, the derivative is 0 (no error, hence no change needed).

1. Binary Cross-Entropy:

$$J = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1 - \hat{y})}$$

2. Hinge Loss:

$$J = \max(0, 1 - y \cdot \hat{y})$$

$$\frac{\partial J}{\partial \hat{y}} = \begin{cases} 0, & \text{if } y \cdot \hat{y} \geq 1 \\ -y, & \text{otherwise} \end{cases}$$

2. Kullback-Leibler Divergence (KL Loss): KL Divergence is **fully probabilistic** by nature.

$$D_{\text{KL}}(P||Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad \text{Another Form: } D_{\text{KL}}(y || \hat{y}) = \sum_i y_i \log \left(\frac{y_i}{\hat{y}_i} \right)$$

- $P(x)$ is the **true distribution** (often from data or labels),
- $Q(x)$ is the **predicted distribution** (e.g. from softmax).

$$D_{\text{KL}}(P||Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) = \sum_x P(x) [\log(P(x)) - \log(Q(x))]$$

$$\frac{\partial D_{\text{KL}}}{\partial Q(x)} = -\frac{P(x)}{Q(x)}$$

Here,

$$\log(x) = \ln(x) = \log_e(x)$$

$$\frac{d}{dx} \log(x) = \frac{1}{x}$$

Integral Calculus for Data Science

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✓ Basic Integration Rules

- Power Rule
- Constant Rule
- Constant Multiple Rule
- Sum and Difference Rule
- Exponential Rule
- Logarithmic Rule

$$\int f(x) dx$$

- $\int x^2 dx$: integrate with respect to x
- $\int t^2 dt$: same function shape, but integrating with respect to t

So, when you integrate:

$$F(x) = x^2 + 5 \Rightarrow F'(x) = 2x$$

$$G(x) = x^2 - 100 \Rightarrow G'(x) = 2x \text{ too!}$$

Why is + C (the constant) added?

$$\int 2x dx = x^2 + C$$

There are infinitely many functions whose derivative is $2x$, and they all differ by a constant.

◆ 1. Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + \underline{C} \quad (\text{for } n \neq -1)$$

Example:

$$\int x^2 dx = \frac{x^3}{3} + C$$

Handwritten red notes:

$$\frac{d}{dx} \left(\frac{x^3}{3} + C \right) = \frac{1}{3} \cdot 3x^2 = x^2$$

◆ 2. Constant Rule

$$\int a \, dx = ax + C$$

Example:

$$\int 5 \, dx = 5x + C$$

Here,

$$\frac{d}{dx}(x + C) = 1$$

$$\int 1 \, dx = x + C$$

◆ 3. Constant Multiple Rule

$$\int a \cdot f(x) dx = a \int f(x) dx$$

Example:

$$\int 3x^2 dx = 3 \int x^2 dx = x^3 + C$$

Here,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

◆ 4. Sum/Difference Rule

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

Example:

$$\int (x^2 + 2x) dx = \int x^2 dx + \int 2x dx = \frac{x^3}{3} + x^2 + C$$

Here,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

◆ 5. Exponential Rule

Example:

$$\int e^x dx = e^x + C$$

Here,

$$\frac{d}{dx}(e^x) = e^x$$

$$\int e^{2x} dx = \frac{1}{2}e^{2x} + C$$

$$\int e^{ax} dx = \frac{1}{a}e^{ax} + C$$



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$$\int e^{2x} dx \quad \xrightarrow{\frac{d}{dx}(u)} \quad \frac{d}{dx}(2x) = 2$$

Let $u = 2x$, so: $\frac{du}{dx} = 2 \Rightarrow dx = \frac{1}{2} du$

Substituting: $\int e^{2x} dx = \int e^u \cdot \frac{1}{2} du = \frac{1}{2} \int e^u du = \frac{1}{2} \cdot e^u + C$

Replace $u=2x$ back: $\frac{1}{2} e^{2x} + C$

Rule to remember: $\int e^{ax} dx = \frac{1}{a} e^{ax} + C$

◆ 6. Log Rule

$$\int \frac{1}{x} dx = \ln |x| + C \quad (\text{for } x \neq 0)$$

Example:

$$\int \frac{1}{3x} dx = \frac{1}{3} \ln |x| + C$$

Because:

$$\frac{d}{dx} \left(\frac{1}{3} \ln |x| + C \right) = \frac{1}{3} \cdot \frac{1}{x} + 0 = \frac{1}{3x}$$

Thank you!